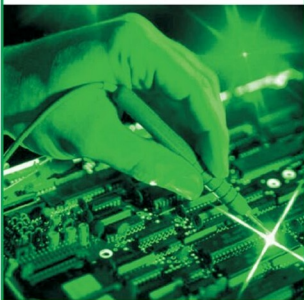


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the sensor is connected to channel 0 (CH0) of ADC IC1 as shown in Fig. 1. IC1 is connected to Arduino controller in SPI mode. The sensor generates voltage output from 0.2V up to 4.8V for 5V power supply. Resolution of ADC IC1 is $5/2^{12} = 1.22\text{mV}$. The input voltage to channel 0 of the ADC IC1 gets converted to equivalent 12-bit binary data and is fetched by Arduino board in two bytes. Board1 then processes the digital data from IC1 and shows it on the serial LCD together with the Pole (North/South) indication.

Software

The software is written in Arduino programming language and the board is programmed using Arduino IDE software. ATmega328 on Arduino Uno board comes pre-burnt with a bootloader that allows you to upload new code without the use of an external hardware programmer. It communicates using the original STK500 protocol. You can also bypass the bootloader and program the microcontroller through the in-circuit serial programming (ICSP) header, but using bootloader, it is quick and easy. Select Arduino Uno from the 'Tools > Board' menu (according to the microcontroller on your board) in the Arduino IDE and burn the program through standard USB port in the computer.

Construction and testing

An actual-size, single-side PCB for Arduino-based Gauss meter is shown in Fig. 2 and its component layout in Fig. 3. Assemble the circuit on the provided PCB to minimise the assembly errors and save time. Use IC base for ADC IC1.

To test the circuit for proper functioning, check 5V power supply for IC1 and IC2 at TP1 with respect to TP0. The analogue voltage levels corresponding to magnetic field strength can be observed at TP2 using a multimeter. ●

The author is a B.Tech in electronics and communication engineering

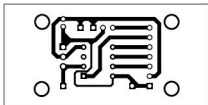


Fig. 2: An actual size, single-side PCB for the Gauss meter

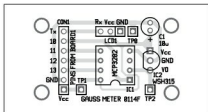


Fig. 3: Component layout for the PCB

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on source.efymag.com website.